

1. EXECUTIVE SUMMARY & STRATEGIC VALUE

Under the National Smart Grid Mission and RDSS guidelines, CESC is ingesting millions of 15-minute smart meter readings across Kolkata and Howrah. Processing high-frequency telemetry, load balancing, and theft detection on public cloud platforms (AWS/Azure) creates two fundamental challenges: **escalating annual cloud compute expenditures** and **regulatory exposure under Central Electricity Authority (CEA) / CERT-In critical power grid cybersecurity mandates**.

FlockML provides a zero-CapEx, software-only edge execution engine that coordinates CESC’s existing substation SCADA terminals, distribution hub PCs, and billing office workstations into a private, on-premises compute grid—processing all grid telemetry locally at a **~70% cost reduction** with zero data egress.

~72.4%

COMPUTE COST REDUCTION VS. AWS CLOUD

100% On-Soil

CEA & CERT-IN AIR-GAPPED GRID COMPLIANCE

2. THE ENERGY-COMPUTE ARBITRAGE FOR RPSG GROUP

Globally, hyperscalers (Amazon, Microsoft) are investing billions into power generation because electricity constitutes the largest operating expense in AI computing. CESC possesses the foundational assets: **raw power at generation cost (₹3–4/kWh)**, **substation real estate**, and **established dark fiber right-of-way**.

RPSG Subsidiary	FlockML Application	Quantifiable Value
CESC Power Grid	Substation smart meter analytics, transformer thermal overload detection, and peak load forecasting.	Cuts ~₹1.48 Cr/year in recurring cloud inference costs; eliminates DSM grid deviation penalties.
Firstsource (BPO)	Local LLM inference for customer interaction automation & speech transcription.	Reduces third-party OpenAI/Anthropic API costs by ~70% across global contact centers.
Woodlands Hospital	On-premises medical image diagnostics and patient record NLP.	100% patient data privacy within Kolkata; zero cloud leakage.

FLOCKML x CESC: 14-DAY SUBSTATION PILOT

Zero-Risk Technical Validation & Implementation Scope

PILOT PROPOSAL

3. ARCHITECTURE & ZERO-RISK ISOLATION GUARANTEES

The FlockML edge engine is designed with strict **IT/OT network isolation**. It operates purely as a passive telemetry consumer without any write-access to electrical breakers or SCADA actuation controls.

Zero-Interference Operational Guarantee:

- Read-Only Telemetry Ingestion:** Connects to read-only MQTT/Kafka smart meter mirrors or CSV data feeds.
- Sandboxed Execution:** Runs in lightweight WebGPU/WASM browser workers without requiring kernel drivers or OS modifications.
- Sub-5ms Work-Stealing Failover:** If any substation node loses power or connectivity, remaining cluster nodes automatically assume the workload without packet loss.

4. 14-DAY PILOT SCOPE & DELIVERABLES

Phase	Timeline	Objective & Action Item
Phase 1: Setup	Days 1 – 3	Deploy FlockML lightweight worker on 10 idle office/substation PCs across Chowringhee, Salt Lake, and Howrah.
Phase 2: Ingestion	Days 4 – 10	Feed 50,000 anonymized smart meter telemetry streams; benchmark local peak load prediction vs. historical actuals.
Phase 3: Audit	Days 11 – 14	Deliver comprehensive Executive Audit Report: Latency, Cost Comparison vs. Cloud GPU, and CEA Compliance Report.

5. NEXT STEPS

We propose scheduling a 30-minute technical scoping session with the **Chief Technology Officer (CTO)** or **Head of Grid Operations** to review network boundaries and approve the 10-node pilot protocol.

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